BASE CHANGE COMPARISON FOR THE RIGHT ADJOINT COMPOSED WITH PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0A9K (sources/stacks-duality/pinned-original-source.tex line 957).

Allowed standard prerequisites (editorial).

- Base change for the derived pushforward. Derived Categories of Schemes, Lemma labelled perfect-lemma-compare-base-change, cited at proof lines 963-967. Gives the isomorphism of the left adjoints.

Source reference. Original equation 0A9J: sources/stacks-duality/pinned-original-source.tex, source lines 927–927. Same-chapter pointer into the author environment equation at source line 927; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08IB: sources/stacks-chow/tags.txt, source lines 10809–10809. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 967 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *In diagram (0A9J) assume that g is flat or more generally that f and g are Tor independent. Then $a \circ Rg_* \leftarrow Rg'_* \circ a'$ is an isomorphism.*

Proof. In this case the base change map $Lg^* \circ Rf_*K \longrightarrow Rf'_* \circ L(g')^*K$ is an isomorphism for every K in $D_{QCoh}(\mathcal{O}_X)$ by Derived Categories of Schemes, Lemma 08IB. Thus the corresponding transformation between adjoint functors is an isomorphism as well. □

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